

all TE pairs. Similarly, paraphrase programs can be either recognisers, generators or extractors. Before we look into the techniques for TE and paraphrasing recognition/generation/extraction, let us look at the practical applications for TE and paraphrasing in NLP applications.

Applications of paraphrasing and textual entailment

So far, we have been considering textual statements as inputs to paraphrasing systems. But this may not always be the case. In fact, one of the earliest NLP applications of paraphrasing was in the field of automatic ‘Question Answering’ (QA) systems. Consider a QA system in which the system is given a set of documents, and needs to find the answer to the posed question from among the documents. Given that the answer, if it is indeed present in the document, may be phrased differently from the way the question has been framed, it may be necessary to generate paraphrases of the question and check if any of the paraphrased questions can be answered using the content in the document collection. Let us consider a simple example in which the question being posed to the QA system is, “Who is the author of the book ‘Crime and Punishment’?” and one of the documents in the collection contains the passage, “Some of Leo Tolstoy’s finest works include ‘Crime and Punishment’ and ‘Anna Karenina’.” In this case, by paraphrasing the question as, “Whose work is ‘Crime and Punishment’?” the QA system may be able to easily figure out the answer as ‘Leo Tolstoy’. Instead of paraphrasing the question, in some cases, the possible answers may also be paraphrased to check if any of the paraphrases can serve as an answer to the question.

Text summarisation is another area where paraphrasing techniques are used widely. Given a set of text documents to be summarised, one of the important functions of a summary extractor is to identify the most important sentences from the texts to be summarised. For example, let us consider that the task given is to create a summary from all news articles in the last one month which discuss the Ebola virus epidemic. Since these newspaper articles are discussing the same event, many of the documents will contain sentences that convey almost the same information. Hence, it is important to avoid selecting sentences that are paraphrases in the summary. Hence, paraphrasing techniques can be applied to a particular sentence to check if it is a paraphrase of any of the existing sentences in the summary, and if found to be so, it can be discarded. Similar to paraphrasing, TE

can also be applied on a particular statement to check if it can be inferred from any of the existing sentences in the summary. And if so, then the statement can be discarded. Note that paraphrasing can also be used to achieve sentence compression, since it can help to generate a sentence which is shorter than the original sentence but conveys the same information.

Machine translation is another major area where paraphrasing and textual entailment techniques are applied. Paraphrasing, in particular, has been widely used in training machine translation systems by using a human generated translation reference and checking to see if the machine generated translation is a paraphrase of the human generated one. This is typically useful when popular literary works, which have been translated by humans, are used to train the machine translation systems. Let us consider a simple example, where the two statements are:

(a) Company A filed a case against Company B for copyright infringement.

(b) Company A accused Company B for copyright violation.

If our machine translation system has never encountered the phrase ‘filed a case’ in the source language during its training samples, it could try finding a translation for the paraphrased source sentence (b), if it has encountered the word ‘accused’ before in its training samples. Using paraphrasing allows a machine translation system to deal with words that it has not encountered before in its training samples.

Other areas where TE and paraphrasing techniques are useful include Information Extraction (IE) systems, which use automated techniques to extract information on specified topics from various sources. These are typically used for answering natural language queries in information retrieval systems. Consider, for example, a system that extracts information on all motor vehicle accidents from newspaper reports. Consider the following statement:

(a) The tourist bus collided with an oncoming car and crashed on to the median.

If this can be paraphrased as “There was an accident involving a bus and car,” then it is possible for the IE system to include this as a candidate news item in its search. Other areas in which TE and paraphrasing can be applied include automatic grading of students’ answers, and in providing simplified summaries of the expert documents in a form understandable to laymen.

We will continue our discussion on TE and paraphrasing next month, when we will look at the algorithms for them. Meanwhile, I would like to give readers the following assignment.

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